

seeds from the air have also been invented, which could help reforestation efforts in places that have been clear cut.

Construction. Mapping and surveying construction sites can be quite slow when done by walking a site. Drones help speed up these efforts, allowing construction companies to provide real-time maps of progress and surveys that can help them both plan for projects and improve projects that are underway, leading to significant savings.

Delivery. Consumer drone delivery has yet to be rolled out at a large scale anywhere in the world but it does present a major contribution for commercial drones. Medical drone delivery is currently taking place throughout the world in countries as far-reaching as Rwanda, the US, and Switzerland (where Flyability is headquartered).

Filmmaking. For years now, high-end drones have been used instead of helicopters to capture aerial shots for movies as latter are more expensive and cumbersome to work with.

Mining. Mining companies are turning to tough indoor drones like the Elios 2 to help them create maps of their mines. These maps lead to improved safety and can also help companies locate ore that might otherwise be lost.

Insurance. Insurance companies are always processing claims, especially after large storms. Drones are helping insurance companies process claims on roof damage much more quickly by allowing adjusters to collect visual data from the sky instead of by climbing up ladders. Insurance companies are also using drones for accident reconstruction, helping them to piece together how an auto collision took place so that they can verify the validity of auto-related insurance claims.

Oil & Gas. Indoor drones like the Elios 2 are making a big impact in oil & gas industry by providing inspectors with a tool for collecting high-quality visual data inside assets crucial to the oil refining process such as tanks and

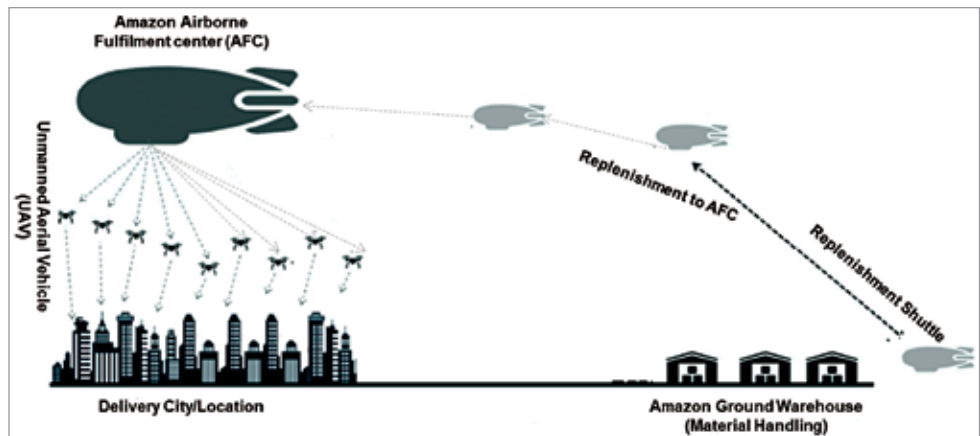


Fig. 2: Implementation design of Amazon's flying warehouse

FCC units, and risers.

Power generation. In power generation, indoor drones are also helping inspectors to access areas that would otherwise be difficult to reach. Drones can also help keep inspectors away from the possibility of harm by radiation at nuclear power plants by taking the place of inspectors in collecting visual data of key assets like boilers.

Public safety. Law enforcement, fire departments, and search and rescue have all adopted drones over the last several years. Police use drones to help them get better situational awareness and to map densely populated areas, firefighters use drones to collect thermal data that can pinpoint where they should focus their efforts, and search and rescue personnel are using both thermal and visual sensors on drones to help find people missing in the wilderness.

Sewer maintenance. Indoor drones have been helping inspectors enter city sewer systems to collect visual data that can be used to identify the source of a problem or to evaluate the condition of the infrastructure as part of the regular maintenance process.

Amazon's flying warehouse - technical aspects

American e-commerce giant Amazon will fly a warehouse at an altitude of around 13,700 metres (45,000 feet) that would act as a launch pad for UAVs, with an aim to deliver consignments in a jiffy, media reports announced recently. Amazon, after winning a patent for flying the ware-

house in the US, announced plans for an airborne fulfilment centre (AFC) such as an airship.

The airship will be stocked with lots of products and would float at an altitude of around 13,700 metres, it said. When a customer places an order, a drone will fly down and deliver the parcels. Amazon said the process would require little power as the drone will be gliding down rather than having to take off and land.

"When the UAV departs the AFC, it may descend from the high altitude of the AFC using little or no power other than to guide the UAV towards its delivery destination and/or to stabilise the UAV as it descends," the patent filing explains. The firm revealed several uses for the flying warehouse.

One example is that of a football match where customers may want certain items such as food or merchandise. Ahead of the game, the AFC could stock up on items and deliver these during the game through UAVs when ordered. It could also be used as a giant advertising board, allowing customers to order the items on display, all of which can be ordered within minutes.

The UAVs would be able to communicate with each other via a mesh network to give information such as weather and the best route for driving. These could also recharge on the airship, as per the report. Amazon's filing explains that the blimp would remain in the air and be refuelled and replenished using a shuttle. This could be a smaller aircraft capable of docking onto the AFC and unloading products as well as fuel.